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## ***Dietary Guidelines in Chronic Liver Disease***

Diagnosis: NASH - Cirrhosis of liver, Child A, Obese, Sex: M  
Type 2 DM

Current Dry Body weight: 92.7kgs

BMI: 34.9kg/m<sup>2</sup>

Total calorie recommended: 1600kcal

Total protein: 90g

### **General guidelines**

- Always eat and drink from hygienic and clean sources to prevent getting infections.
- A high protein and low salt diet is to be taken.
- **Divide your meals into 6 to 7 times per day and take small frequent meals with a bedtime snacks not to be missed**
- You can take animal protein like egg, fish and chicken though red meat is restricted. If possible vegetarian protein is preferred as it has fibres and would help into keeping the bowels clear which is important in cirrhosis.
- While cooking you can use turmeric (haldi), spices without salt, limited amount of oil
- **Spend at least 15-20mins every day in sunlight for Vitamin D.**

**Carbohydrate intake:** It is preferable to avoid simple sugars and take more of complex carbohydrates

| <b><i>Preferred carbohydrates</i></b>                                                                                                                                | <b><i>To avoid carbohydrate</i></b>                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><i>Complex Carbohydrates:</i></b> Peas, beans, whole grains, vegetables, fruits (apples, banana), oat meals, hand pounded or single polished rice, jowar, millets | <b><i>Simple carbohydrates:</i></b> Avoid refined flour – maida, white bread, bakery products likes cakes, pastries, puffs, cookies, doughnuts, chips, aerated and Fizz based drinks (coke, pepsi, sprite etc). Table sugar, corn syrup |

**Proteins:** Since the liver is not able to produce protein, we need to supplement proteins from outside. Though vegetarian protein is preferred, people can take non-veg proteins also. We should target 1.2 to 1.5 g/kg/body weight. This will include protein from regular food and supplements.

| <i><b>Vegetarian source of protein</b></i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <i><b>Non-vegetarian source of protein</b></i>                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Lentils (each cup 18g) – Indian Dal: Masoor dal has highest protein</li><li>• Legumes (each cup 15g) – kidney beans, black beans, chick peas</li><li>• Nuts (15-20 per day): Almonds, cashew nuts</li><li>• Green peas (9g/cup)</li><li>• Oats (6g/cup)</li><li>• Quinoa, sprouts, channa</li><li>• Chia seeds (35g)</li><li>• Soya milk (7g/cup)</li><li>• Fruits like banana, guava, berries</li><li>• Greek Yogurt</li><li>• Tofu</li><li>• Soya and soya chunks</li><li>• Spinach</li></ul> | <ul style="list-style-type: none"><li>• Skinless chicken</li><li>• Egg white</li><li>• Fish especially salmons, mackerel, tuna</li></ul> |

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## Diabetes Dietary Recommendation (in case you are also diabetic)

- Diet should be rich in fruits, vegetables, whole grains, whole gram pulses, & sprouts to provide for vitamins, minerals, antioxidants & fibre
- Meal timings and consistency in composition of meals: Keeping the number of meals, scheduled time for eating & the composition of meals must be relatively constant from day to day. **Small, frequent meals help to prevent large glucose excursions and curtail hyperglycaemia.**
- Person with NIDDM are at an increased risk of premature atherosclerosis and it is recommended to **limit fat and cholesterol intake.**
- Restriction of simple carbohydrates: Simple CHO such as sucrose, glucose, fructose need to be restricted with inclusion of diets rich in complex CHO (such as whole wheat, jowar, corn, hand-pounded rice) and fibre.
- **FIBRE:** Studies have suggested that intake of soluble dietary fibre leads to decrease in plasma glucose and reduction in insulin requirement. It is **recommended to increase the dietary fibre intake.**

## Artificial Sweeteners

Non-calorie sweeteners (aspartame & saccharin): The use of these sweeteners is acceptable for people with diabetes when used within the FDAs safety levels.

- The recommended safe limit for saccharin: **1000mg/day for adults (0.2tsp).**
- The recommended safe limit for aspartame: **50mg/kg body weight /day**
- The use of artificial sweeteners during pregnancy is **NOT** recommended.

| <i>Fruits to be avoided X</i>                  | <i>Fruits to be included ✓</i>                           |
|------------------------------------------------|----------------------------------------------------------|
| Mango, Banana, Custard Apple, Chickoo & Grapes | Rest all fruits can be taken, but <b><u>NO juice</u></b> |

## Estimate protein in food

|                                                     |       |
|-----------------------------------------------------|-------|
| Chicken approx. size of deck of cards cooked        | 20 gm |
| Fish (baked/fried/steamed) of size of deck of cards | 18 gm |
| Tofu of 150 gms                                     | 21 gm |
| Chick peas, beans, lentils 3/4th cup                | 11 gm |
| Peanut, almond unsalted                             | 7 gm  |
| Milk skim -1 cup                                    | 9 gm  |
| Greek yogurt 3/4th cup                              | 16 gm |
| Cheese                                              | 10 gm |
| Whole wheat bread                                   | 5 gm  |
| Egg large                                           | 6 gm  |

**Fat intake:** 0-20% of calories should come from fat, with an emphasis on intake of mono- and polyunsaturated fat and minimization of trans and saturated fat. Modified dietary fat as MCT oil is sometimes necessary. **Oil and ghee can be taken in limited amount and it is not required to consume boiled food**

**Sodium intake:** Restricting sodium in diet is the key to reducing leg swelling and preventing water accumulation in the abdomen. Major source of sodium is salt. But, we should look for hidden salts which are present in many food. **Limit sodium intake 2g per day. (½ tsp) which is equal to 5g of salt = 1tea spoon of common salt)**

**DO NOT put salt while cooking. Please keep aside 1 tsf salt for the patient separately, which the patient will sprinkle over his food over 24 hours**

| Hidden source of salt ( Avoid ) ✗                                                                                                                                                                                                                                                                   | How can you substitute lack of salt in food ( can take ) ✓                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>Bakery products which contain sodium in baking powder eg. Breads, puffs, do noughts etc</li><li>Soya sauce and other sauces</li><li>Canned and frozen food</li><li>Salted butter, Achaar, Papad</li><li>Salted cheese, vinegar, Puffed rice (Moori)</li></ul> | <ul style="list-style-type: none"><li>Use home grounded black pepper</li><li>Coriander leaves</li><li>Basil</li><li>Ginger and garlic</li><li>Dry mustard</li><li>Onion</li><li>Mint</li><li>Lemon juice</li></ul> |

(The original page also has a few handwritten Bengali notes in the margin of this section — too unclear in the photo to transcribe reliably; please check the original photo for those.)

**Potassium:** Potassium needs to be restricted or increased based on the serum potassium levels. Hence monitoring serum electrolytes becomes very essential in cirrhotic patients and according the intake of these foods needs regulation.

**Good sources of potassium: coconut water, fruit juices, cocoa, jaggery, coffee, cornflakes**

**Fluids:** Fluids need to be restricted only if the Serum Na (sodium) - <125mg/dl. Monitor Sr. sodium every 2 weeks if you are on water pills. **Please remember that drinking fluids do not cause water accumulation**

**Avoid Alcohol, smoking and any other addiction.**

**Department of Hepatology: Nutrition: AIG Hospitals: 1600kcal, Salt restricted, high Protein, High fibre diet**

**High fibre diet**

| Time (Flexible)       | Meal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7am (After waking up) | Green Tea/ Black coffee – 1cup<br>With, Soaked almonds- 5pcs, walnuts -3pcs each <b>Or Assorted nuts -30g</b>                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 9am (Breakfast)       | 2pcs- Roti (without oil) (Preferred jowar, bajra, ragi or multigrain) with either mixed vegetables curry (1cup) or cup of cooked beans or peas or Dal<br><b>Or Veg rolled Oats/ Veg Quinoa /Veg dhaliya/ Veg poha/ Veg upma/ Veg khichdi/ Veg Millets – 1 ½ cups with glass of Buttermilk</b><br><b>Or Moong dal chilla/ Besan chilla/ Oats chilla/ Oats Idli/ Dosa -2pcs with Sambar with vegetables</b><br>Or 2-slices of Brown bread with 1cup Museli or corn flakes or ragi flakes<br><b>Plus, Egg –boiled: 1 with 2 pcs Egg whites</b> |
| 10-11am (Mid-morning) | Any Seasonal fruit-1 (100-150g)/ Black coffee -1cup with Assorted nuts – 20g<br><b>OR</b><br>Buttermilk- 150ml +1tsp (Powdered flax seeds + chia seeds + watermelon seeds + sunflower seeds)                                                                                                                                                                                                                                                                                                                                                |
| 1-2pm (Lunch)         | Cleanly washed salad with less salt<br><b>PLUS, either Brown Rice or Lemon rice or Veg Pulao (1katori)</b><br>Or Roti (without oil) - (jowar or bajra or multi –millet) (1pc)<br><b>PLUS, cooked vegetables (1cup size) And Dal (1cup at least)</b><br><b>with either</b> Low fat curd/ Paneer/ Tofu/ Soy chunks/ Soy granules – 1cup<br>OR size of playing card Fish (Steam/ grilled/ baked/ curry/ Boiled)/ Skinless (Country) chicken or Egg whites –2pcs                                                                                |
| 4pm (Teatime)         | Green tea/ Black coffee – 1cup with protein bar - 10g                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 6pm (Mid Evening)     | Either Fruits-1cup (especially berries & citrus fruits)<br>Or Sprouts (steamed)/ Chenna/ Boiled peanuts/ Boiled corn –1cup or Paneer tikka -1cup                                                                                                                                                                                                                                                                                                                                                                                            |
| 8pm (Dinner)          | Cucumber raitha/ Veg soup -1cup<br>Roti (without oil) - Preferred Jowar or Bajra or Multi–millet (1pc)<br>with Boiled white beans with garlic & cumin – 1cup<br><b>with either</b> low fat curd/ Paneer/ Tofu/ Soy chunks/ Soy granules – 1cup<br>Or Egg whites – 2pcs                                                                                                                                                                                                                                                                      |
| 10pm (Bedtime)        | Buttermilk–150ml with Oat bar/ Assorted nuts/ Unsalted peanuts- 20g                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

**Nutritional Supplements: (incl. BCAA)**

**Lisoval Sachet -1Sachet +150ml water (Post exercise daily - 6months)**